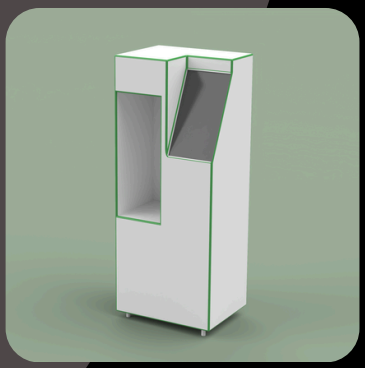
TECHNICAL PORTFOLIO

Alex Fu



tymewear





B.S. in Mechanical Engineering
Minor in Biomedical Engineering
Class of 2029

My interests lie in Biomechanical and Clinical Engineering, bridging MechE with medicine and physiology.

Engineering Intern

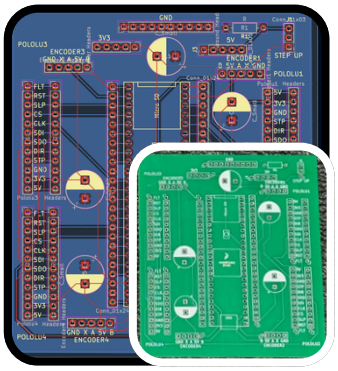
Summer 2026



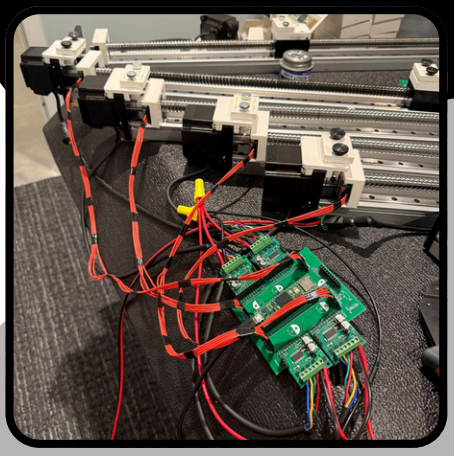


tymewear

Custom Circuit board

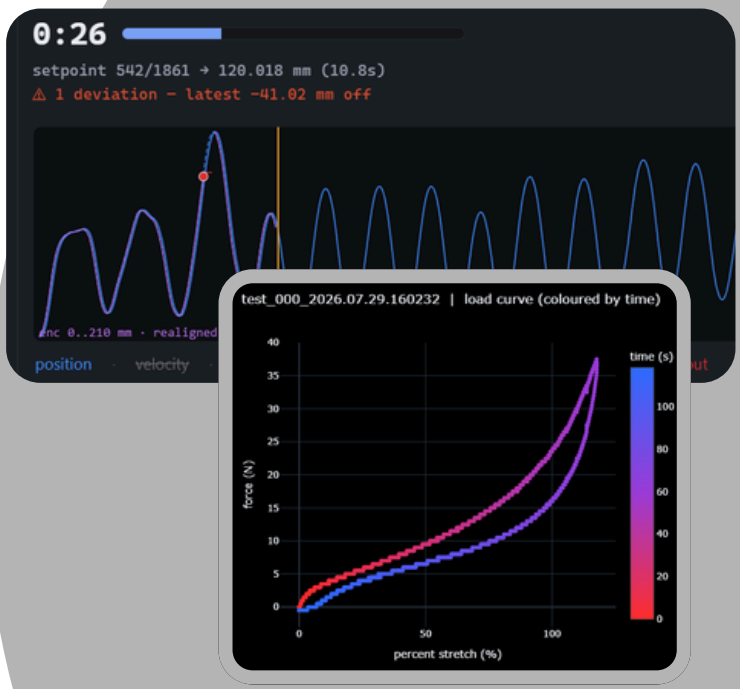


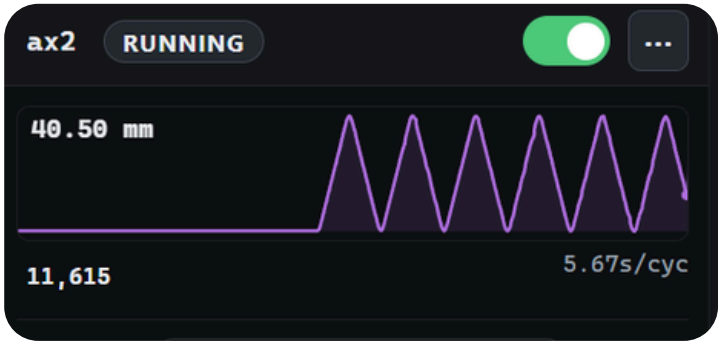
- I designed a custom PCB in KiCAD to replace the jumper wiring my system used, consolidating the whole rig onto one board so it could run without connections loosening
- It includes optimized wiring (no sharp angles, signals away from strong voltage, etc)



Product Testing

- The rig needed to run unattended for long durability runs, but also let someone configure and inspect a single strap in detail. I built the system on a Raspberry Pi 5 and a web interface that loads a protocol, launches it, and logs every cycle
  - **C++ (Teensy firmware):** controls motors, with microstep throttling to dramatically reduce audible noise and spinouts under fast speeds, increasing the speed cap from 50 mm/s to 250 mm/s
  - **Python + Flask (backend):** headless pipelines that run on the Pi, simultaneous motor-encoder-strap aligned recordings to compare inputs to outputs, making data analysis possible from both visual graphs and organized spreadsheet
  - **HTML, CSS, JavaScript (front end):** live position and load plots during stepped force tests and re-running past breathing data, cycle counter, and a deviation alarm that fires when the strap drifts off setpoint mid-run





- Documented the iterations and maintenance of the product testing system using AI:
  - Versioned per-module READMEs so a new session could pick up the existing architecture accurately instead of hallucinating false ones
  - Constrained changes to extending existing working code rather than rewriting them, which kept errors contained to the module I was touching
  - Automated headless data pipelines were used on the Pi rather than need repeated AI use

LymphSense

Freshman Year

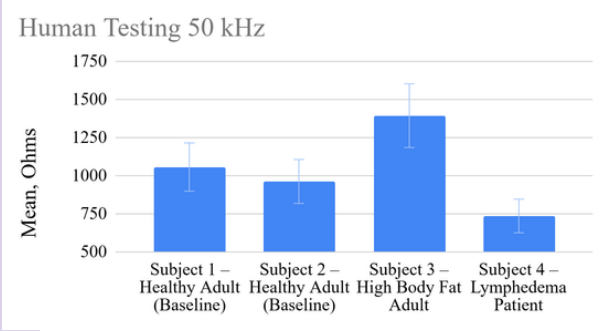


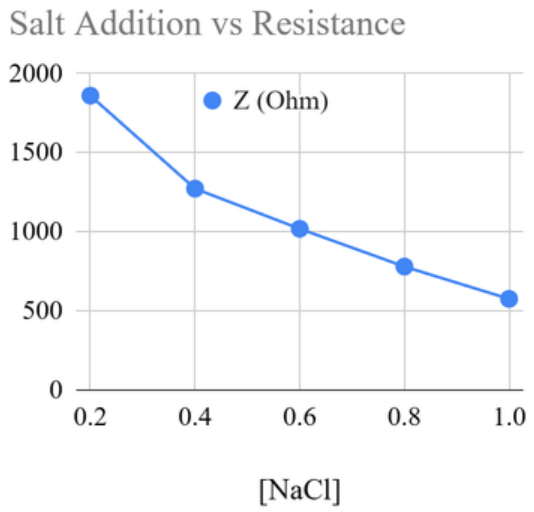


Lymphedema affect 10 million people in the USA alone. Only early stages of lymphedema are reversible, but hard to catch, many not opting to monitor high-risk areas until it's too late.

To make local lymphedema monitoring more realistically viable to incorporate into patients' lives, we've created a low-cost, portable, handheld bioimpedance spectroscopy device, LymphSense. It's designed to measure local fluid changes in areas at risk for lymphedema, poised to be used at home daily.

Our design won 1st place + Student Choice in the year-long BMES Design-a-thon, and Best Junior Entry for BU's Dean's Imagineering Contest.





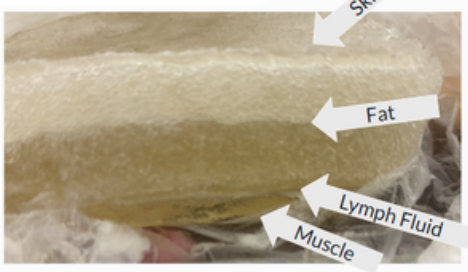
Repeatability Test

(p = 0.01, 50 kHz, 400 mL water, 0.5 tbsp salt)

Z (Ohm)
1269.22
969.28
1116.03
1084.65
838.35
969.38
767.65
1001.81
997.51
1039.58

Mean ± Std
1005 ± 140 Ω

Phantom Testing



What we did

- Reviewed the clinical devices (SOZO, L-Dex, LymphScanner) and DIY builds to find gap
- Interviewed Jennifer Williams, Co-Executive Director of the National Lymphedema Network. Created framing: helping doctors gather data between appointments
- Designed housing and electronics for actual manufacturing, including ideas for volume discounts and material substitutions
- Mapped the regulatory path (LymphSense is Class II, 510(k), SOZO as predicate)

Testing

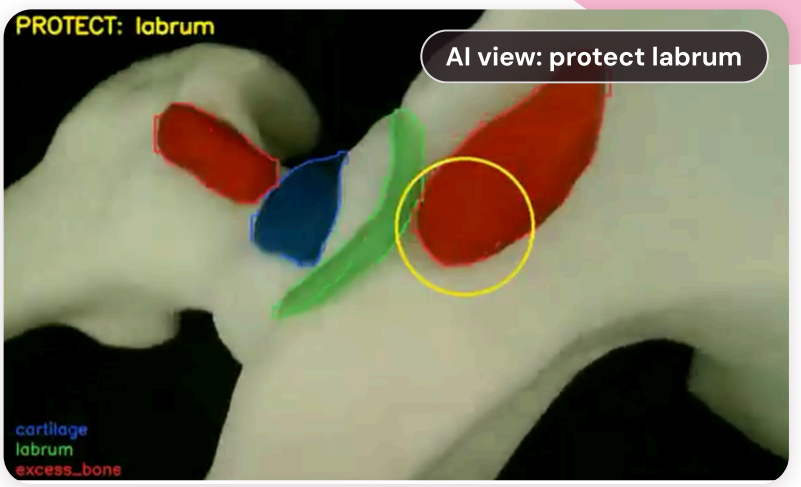
- Saline: impedance dropped as salt went up, 1860 Ω to 576 Ω. Proves it tracks conductivity
- Repeatability: 1005 ± 140 Ω over 10 readings. Looser than we technically want
- Phantom: four-layer skin/fat/muscle/lymph models, one with 20% more fluid. 1075 Ω vs 784 Ω
- Human: four subjects, right leg, 50 kHz. Lymphedema at 735 Ω, healthy at 961–1055 Ω, high body fat at 1393 Ω

TissueGuard

Sophomore Year

MOONSHOT × ADVAMED
NextGen Robotics Competition
MedTech Conference · Boston · Oct 2026



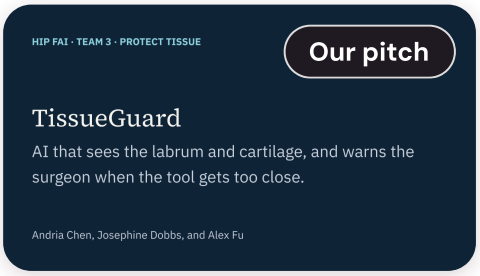


Introducing TissueGuard, a surgical AI robotics tool poised to help doctors protect critical internal structures they designate themselves, creating a more confident doctor.

In a 6-hour designathon, our trio created an AI computer vision monitor that warns the surgeon with non-distracting notifications when the tool gets too close.

Tyler, 25, is a recreational soccer player with shooting pain in his inner groin. Ibuprofen stopped helping as the pain got worse.





Built with Andria Chen and Josephine Dobbs, using Moonshot Forge, Roboflow and the provided robot

What I did

- Mimicked clinical workflow, from **questioning the patient** and **prescribing imaging**, to analysis and AI model training
- Modeled hip structures referencing **X-Ray and CT scans** of a patient with FAI (Femoroacetabular Impingement)
- Trained competing models in **Roboflow** and chose **YOLO26** as the most accurate and efficient
  - Identifies **labrum**, **cartilage**, **excess bone** and **tool tip**
  - 75-20-5 split on **80** labeled images
- Presented our demo and pitch to medtech executives from **Stryker**, **Zimmer Biomet** and others

80	4	100%	96%
labeled images	test images	test mAP, structures	test mAP, tools

With a small test sample size of 4 images, the results are promising, but further testing and training are needed.

BetterYears

Sophomore Year

SUNDAI HACK 143
Biomarkers of Aging
Harvard · Oct 4, 2026





I created minigames with the Apple Vision Pro to calculate your 'BetterYears Age'.

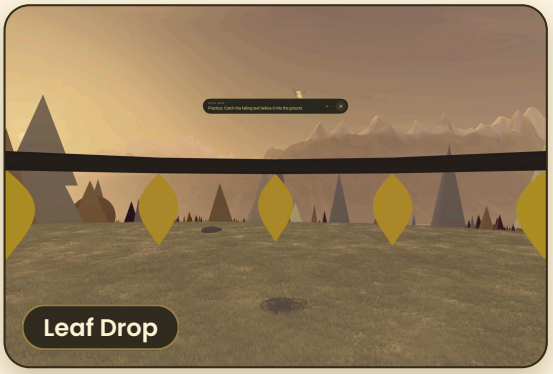
As part of a 10 hour hackathon, I worked on a diverse team of engineers, data scientists, AI experts, and business professionals to create BetterYears. This uses engaging minigames to see a person's longitudinal progress, tracking their reaction time, useful field of vision (UFOV), mobility, stability, and other biometric markers of health to calculate a relative score.

What I did

- I wrote the **Games Ideas deck**: **5** games, each aimed at one thing the body does every day
  - **Leaf Drop**, **Spatial Tracking**: reaction time to what you see and hear
  - **Spatial Memory**: decision time and spatial memory
  - **Scary Balance**, **Hole in the Wall**: reach, holding still, arm mobility
- I designed for **older, less steady players**
  - No overly risky behaviors, such as blind movement, and every walk spot sits within **0.8 m** of the start
  - Hole in the Wall cutouts scale to each player's height (stature = eye height / **0.936**)
  - Blue = touch, orange = leave, and orange targets change shape so color is never the only cue
  - Optional practice round before every game, one tap to skip
- I defined the **Dusk look**: charcoal glass, cream text, peach accent, never bright white
- I built **Dusk Meadow**, the 3D forest behind every game. Only the nearest ground, trees and grass are real 3D (**~470k** triangles, **500k** budget)
- I made the soundtrack with Suno



Built with Hunter Harris, Jason Morris, Jessica Myles, Franco Miguel Valencia, Ben Weisburd and Wilson Wu





SeedSwap

Freshman Year

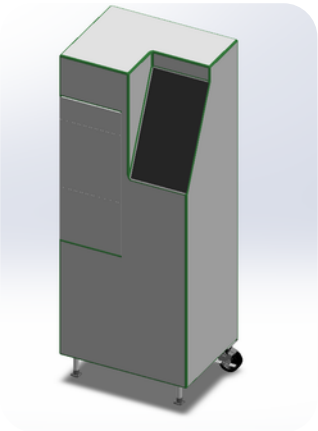
I designed SeedSwap and won first in the Ripple Design-a-thon, a 40 hour competition hosted by MIT Design for America and Wellesley In-Product.

Our team won First in the Physical Track, awarded for strong emphasis on physical usability; and Second in the Design For America (DFA) Track.

We created a vending-machine styled seed sharing device. You interact with this screen interface, optionally set up an account, and select which seeds you're inputting or taking. You don't have to swap; you can donate or just take a few.



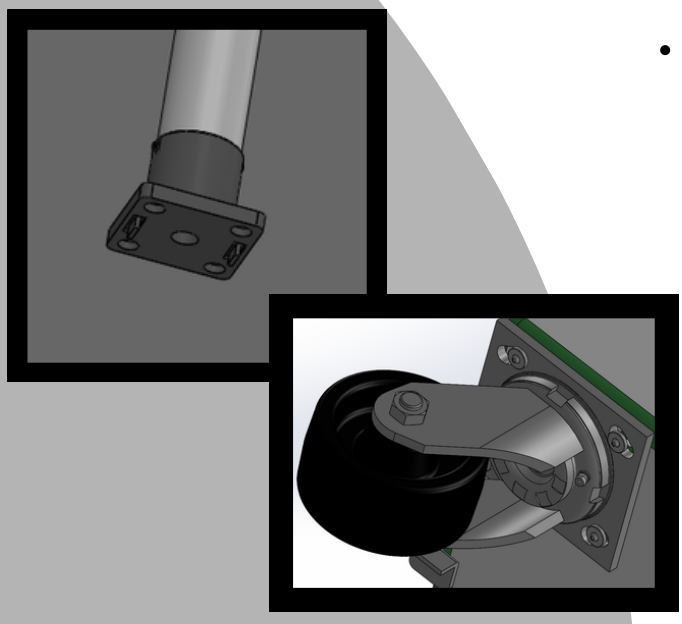
Bin Demonstration





I was the mechanical engineer in our team of three, focusing on **SolidWorks's 3D Model, Assembly, and Visualizer**

- I researched **human-centered engineering principles** to incorporate into the design. Examples
  - a 25° angled screen to reduce screen glare and neck strain
  - bin heights that would accommodate 95% of the population



Hybrid Feet/Wheels for Portability + Stability

- **I included mechanisms for**
  - kiosk stability
    - adjustable and modular leg layout
  - visual bin cues for intuitivity + elderly accessbility
    - high contrast colors + LEDs
  - robust material selection
    - Powder-coated 16-gauge steel for longevity in outdoor environments
    - Closed-cell polyurethane foam (~1–2" thick) for seed insulation
  - climate control
    - Temperature through TMP36 + thermoelectric cooler
    - Humidity through Hygrometer Detection Module Moisture Sensor FC-28 DC 3.3V-5V I1F67124